

“The next five years are expected to see more technology inflections and shifts than the past 15 years”



As we advance from one generation of technology to the next and develop smaller, faster and more functional chips, there are key inflection points at which innovative technologies must be developed. This interview discusses the key emerging technologies in the semiconductor space, including an alternative to silicon itself.

Dr. Randhir Thakur has over 300 patents to his name and has published more than 200 papers. He was named an IEEE Fellow in April 2013 for his ground-breaking leadership in the development and implementation of single-wafer technology in semiconductor manufacturing. In an interview, Abhishek Mutha of EFY talks to Dr. Thakur, executive vice president and general manager of the Silicon Systems Group at Applied Materials Inc., about the inflection points in today's semiconductor industry and the evolution of the process to make microchips.

DR RANDHIR THAKUR

EXECUTIVE VICE PRESIDENT AND GENERAL MANAGER OF THE SILICON SYSTEMS GROUP AT APPLIED MATERIALS INC.

What are the key inflection points that designers should be aware of?

As the industry moves from one technology generation to the next to make smaller, faster and more functional chips, there are key inflection points where innovative technologies must be developed to meet new manufacturing challenges.

The first is transistors, which are effectively the 'on-off' switches of each device. They are incredibly complex and so small that billions of transistors are packed into a microchip. And they are getting even more complex and smaller. However, to make these transistors work reliably at the more advanced technology nodes, new materials and designs are needed.

The next area is 'patterning.' To pack billions of transistors into today's most advanced microchips, manufacturers must seemingly defy the laws of physics, using complex lithography methods to print or 'pattern' the wafer with chip features that can be a 1000 times smaller than a human hair.

Then there are the 'interconnects,' which are the electrical pathways that connect transistors and other circuit com-

ponents to one another and to the outside world. They are critical to the speed and reliability of a microchip. Because modern microprocessors can have as many as ten levels of interconnects, this intricate structure is one of the most process-intensive—thus most costly—part of the total chip fabrication process.

Packaging is another area where inflections are being encountered. An emerging area for Applied Materials' technology is in three-dimensional integrated circuits (3D-ICs). In a 3D-IC, multiple chips are vertically stacked in a single package to deliver higher performance and functionality in a smaller area. The chips are electrically connected using deep holes called through-silicon vias (TSVs).

3D NAND memory also poses a challenge. After following Moore's Law for nearly 40 years, increasing the storage capacity of traditional, two-dimensional memory chips even while decreasing the size of the features on them (called scaling) is becoming very difficult now. For example, an advanced 25nm Flash memory chip stores each bit of information using approximately 100 electrons. Containing those electrons reliably over millions of read/write cycles is a major challenge—and achievement.